



Yaksh Bhesaniya

Backend | Systems | Concurrency & Reliability
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Examination	University	Department	Year	CGPA
Post Graduation	IIT Bombay	CSRE	2027	9.18
Graduation	Gujarat Technological University	Information Technology	2024	8.88
Diploma	Gujarat Technological University	Information Technology	2021	10.00

WORK EXPERIENCE

- **AI Engineer (Freelance) | Dhi Labs AI, Ledgerline; Outlyn (GATE CS & DA ed-tech), team of 4** (Apr'26-Present)
 - Architected the **multi-tenant SaaS backend** (FastAPI, PostgreSQL 16, Redis, S3): per-tenant row scoping, a `SELECT ... FOR UPDATE SKIP LOCKED` worker pool for long-running jobs and an **advisory-lock watchdog** reclaiming jobs orphaned by worker crashes, verified by **528 tests** exercised against real PostgreSQL and Redis rather than mocks, gating every release.
 - Hardened the billing path against retries and duplicate submissions using **24-hour idempotency keys**, per-user **concurrency caps** and a budget guard that terminates a run mid-pipeline while preserving partial artifacts, across **24 migrations**.
 - Owned **Ledgerline's** LLM agent architecture end-to-end: a Discovery/ReAct/Synthesis pipeline answering covenant questions over **350+ page** credit agreements via **23** document tools, capped at **15 iterations**, with full tool-call tracing on every run.
 - Cut client LLM spend per **350-page** agreement (extraction **\$10 to \$1–2**, end-to-end answering **\$5–6 to \$2**) by right-sizing the model per task and caching prompts and artifacts, with **no accuracy regression** on the expert-authored ground-truth set.
 - Designed a **deterministic preprocessing pipeline** (8 LLM stages) indexing raw PDFs into TOC, definitions and cross-references, so the agent reasons over **document structure, not raw text chunks**, with deterministic reruns of the same question.
 - Implemented **multi-provider LLM routing** (OpenAI / Anthropic / Grok) in YAML with prompt caching, automatic provider failover and **Langfuse** cost tracing, so switching provider is a config change, not a code change or a redeploy of any service.
 - Built **Outlyn's** background job queue as a **PostgreSQL table** instead of a separate message broker: workers pick jobs by priority with `SKIP LOCKED`, and jobs stuck after a worker crash are automatically put back in the queue, since jobs are safe to rerun.
 - Added **retries with growing waits** (up to 3 attempts) and alerts only when a job keeps failing, so one flaky job never pages anyone.

Tech: Python, FastAPI, SQLAlchemy, PostgreSQL 16, Redis, S3, Docker Compose, REST APIs, SSE, JWT, RBAC, Concurrency, Idempotency, pytest, mypy, ruff
- **Project Intern | Space Applications Centre (SAC), ISRO, Ahmedabad** (May'26-Jul'26)
 - Contributed to a satellite image mosaicking workflow stitching overlapping high-resolution scenes into a single geometrically corrected mosaic by refining **RPC sensor models** from tie-point measurements in a reproducible processing pipeline.
 - Worked on the image-matching stage, evaluating **deep-learning-based image matching** alongside classical feature-based methods (SIFT/SURF) and comparing the geometric quality achieved by each approach across the same overlapping scenes.
 - Built product generation and review tooling (**Cloud Optimized GeoTIFF** tiled output, quick-looks, GIS footprints and quality reporting), enabling range-request access to imagery too large to load whole, without specialist desktop software.

Tech: Python, PyTorch (TorchScript), OpenCV, GDAL, Rasterio, NumPy, FastAPI, React, RANSAC, Homography Estimation, RPC Sensor Models, COG, Photogrammetry
- **Freelance Software Developer | Self-Employed** (Jun'24-Jun'25)
 - Delivered production backend systems for **retail clients**, owning every stage from requirement analysis through schema design, implementation, code review and deployment, with the client signing off before each release went live to users.
 - Implemented JWT authentication and role-based access control across multiple user tiers, plus inventory, billing and payment workflows on MongoDB and PostgreSQL, with third-party API integrations for payments, messaging and invoicing.

Tech: Node.js, Express.js, MongoDB, Mongoose, PostgreSQL, Redis, REST APIs, JWT, RBAC, Git, Linux
- **Software Engineer Intern | RapidOps Inc.** (Jan'24-May'24)
 - Implemented backend APIs and optimised database queries for production-scale applications, profiling slow paths and refactoring data access to improve response time and reliability under sustained load, with indexes added on the hot query paths.

Tech: Node.js, Express.js, MongoDB, MySQL, REST APIs, Query Optimisation
- **Software Engineer Intern | HumBee Studio** (Feb'23-May'23)
 - Implemented backend modules and features for web applications across the request-handling and data layers.

Tech: JavaScript, Node.js, REST APIs

KEY PROJECTS

- **StreamScribe: Real-Time Scene Captioning over WebRTC | Self-Project** (Jun'26-Aug'26)
 - Built a **3-stage real-time video pipeline** ingesting webcam/screen-share over **WebRTC** (aiortc) and streaming natural-language captions back to the browser over WebSockets at **2–3s end-to-end latency** under sustained streaming on a single GPU.
 - Decoupled the stages with **bounded thread-safe queues**, giving natural **backpressure** and drop-frames-over-lag load shedding so a slow GPU degrades output rate instead of stalling the ingest loop or dropping the session, which keeps the stream alive.
 - Cut vision-language model inference calls **60–70%** with a two-tier deduplication gate: track-ID on **YOLOv8 + ByteTrack**, then cosine similarity (threshold **0.985**) on pooled embeddings, cutting inference cost and latency on every captioning session.
 - Generated post-session narratives via a **MAP-REDUCE LLM orchestrator** with event compression and chunking; only text ever leaves the machine, never raw video frames, giving **privacy by design** on the user's own machine by construction.

Tech: Python, FastAPI, WebRTC (aiortc), PyTorch, YOLOv8, ByteTrack, WebSockets, Multithreading, Backpressure, React, SQLite
- **MobileBizPro: Retail Inventory & Accounting ERP [Code] | Client Project (team of 2), live in production** (May'25-Jun'25)
 - Designed and shipped a multi-location retail ERP backend of **20 data models, 16 REST route groups and 110+ endpoints**, used daily by a mobile retail business for live billing, stock management and end-of-day reconciliation across all its branches.
 - Built an **IMEI-level stock ledger** tracking every handset as a uniquely serialised unit, preventing duplicate sales and giving exact cost-of-goods-sold attribution per sale under concurrent billing sessions across multiple store locations and shifts.
 - Implemented a **double-entry-style money layer** posting every sale, purchase, expense, return and transfer to bank/cash accounts with automatic balance reversal, powering a **14-report** suite with drill-down P&L by product, brand and branch.
 - Cut invoice PDF generation latency **approx. 80%** with a **warm process pool** plus multi-layer caching, and added nightly off-site backups with a one-command restore path, tested end-to-end as part of scheduled disaster-recovery drills before handover.

Tech: Node.js, Express 5, MongoDB, Mongoose, Redis, Caching, Process Pooling, JWT, RBAC, Puppeteer, Nodemailer, node-cron

- **TruckIt: Real-Time Logistics Booking & Fleet Tracking Platform** [Code] | *Course Project, GNR 605: Principles of GIS* (Oct'25-Dec'25)
 - Built a full-stack two-sided freight booking system with separate user and driver portals, exposing **20+ REST endpoints** via Flask Blueprints over a normalised 4-table schema across 40+ locations, with server-side validation on every route and payload.
 - Engineered a multi-stop route optimiser on the OpenRouteService **HGV profile** with a three-tier **ORS to OSRM to Haversine** fallback, keeping routing and quoting available when an upstream provider is down, rate-limited or out of quota.
 - Implemented dynamic pricing over distance, cargo weight, fleet mix and peak-hour multipliers to **1.25x**, plus a **thread-safe** simulation layer (`threading.local` + SQLite WAL) synchronising two concurrent clients in real time without a message broker.
 - Tech: Python, Flask, SQLite (WAL), REST APIs, Multithreading, Graceful Degradation, Leaflet.js, OpenRouteService, OSRM
- **SETU: Spec-Driven Agentic Extraction of Scientific Data** | *M.Tech Thesis under Prof. Pritam Das* (2026-Present)
 - Building **SETU**, a **config-driven LLM extraction pipeline** that turns scientific papers into a **provenance-tracked dataset**, with **hybrid retrieval** (BGE-M3 vectors and SQLite FTS5, fused by reciprocal rank) feeding a self-hosted Qwen model on **vLLM**.
 - Added **three validation gates** before any extracted value is accepted, covering **JSON-schema** checks on the structured output, **unit and dimension** checks with pint, and **grounding** that requires the quote to appear verbatim on the cited source page.
 - Tech: Python, LLM Agents, Tool Calling, Structured Outputs (guided JSON), RAG, Hybrid Retrieval, pint, pydantic, SQLite FTS5, vLLM
- **Landsat Land-Cover Classification & Water-Body Change Detection** [Code] | *Self-Project* (Nov'25)
 - Implemented **K-Means from first principles in NumPy**, with vectorised centroid assignment, empty-cluster reinitialisation and L2-shift convergence, over approx. **1 million** pixel feature vectors, under an explicit no-sckit-learn constraint throughout.
 - Engineered interpretable features (**NDVI, NDWI, brightness**) and a centroid-statistics heuristic for cluster-to-class correspondence, quantifying water-body change across **2005–2025** directly from the classified maps, without any labelled training data.
 - Validated cluster quality against ground-truth land-cover samples and produced change maps quantifying water-body loss and gain across the study period, with per-class accuracy reported for spectrally similar cover types such as wet soil and shadow.
 - Tech: Python, NumPy, Rasterio, Matplotlib, Tkinter, Unsupervised Learning, Remote Sensing
- **GIS Ambulance Routing & Emergency Response Analytics** [Code] | *Self-Project* (Oct'25-Dec'25)
 - Built a full-stack Flask + Leaflet application routing and analysing campus ambulance trips across **18+** geocoded pickup points, with a **3-tier routing fallback** under API-key exhaustion or network failure, backed by a SQLAlchemy schema over SQLite.
 - Implemented spatial analytics: **isochrone service-area zones** (3/5/7-minute) and a route-frequency heat map built from segment-level aggregation of every logged trip, exposed as interactive Leaflet overlays for dispatch and coverage review.
 - Tech: Python, Flask, SQLAlchemy, SQLite, Leaflet.js, OpenStreetMap, OpenRouteService, OSRM
- **Principal Component Transform (PCT) for Multi-Band Imagery** [Code] | *Course Project, GNR 607: Satellite Image Processing* (Oct'25-Nov'25)
 - Implemented **PCA/PCT from scratch** on 11-band Landsat-8 imagery using manual mean-centering, covariance and eigenvector projection without `sklearn.PCA`, with a GUI reporting variance explained and per-top-k reconstruction MSE for each choice.
 - Built an interactive GUI to load multi-band GeoTIFFs, visualise every principal component and reconstruct from the top-k, reporting variance explained per component alongside reconstruction MSE for each top-k choice, making the cost of compression explicit.
 - Tech: Python, NumPy, Tkinter, tiffio, scikit-image, Dimensionality Reduction
- **Multi-Agentic AI for Critical Events Monitoring** [Code] | *Seminar Project, GNR-694* (Jan'26-May'26)
 - Designed a **hierarchical multi-agent architecture** where a Coordinator agent decomposes an earthquake scenario across specialist Infrastructure and Transportation LLM agents and fuses their **schema-constrained JSON** reports into one prioritised, confidence-annotated response plan over a REST API and a CLI runner, so scenarios can be replayed, diffed and compared across model and prompt versions without re-running the agents by hand, making evaluation repeatable.
 - Grounded each agent with a curated **domain knowledge base** (6 topic categories, approx. **50** seismic and urban-vulnerability facts) injected into prompts, reducing LLM hallucination and making every risk classification traceable and auditable.
 - Enforced schema-constrained outputs with parsing and fallback error handling; architecture grounded in a comparative survey of **CAMEL, AutoGen, ReAct and Toolformer** agent frameworks and orchestration patterns before committing to a design.
 - Tech: Node.js, Express.js, Google Gemini API, Multi-Agent Systems, Prompt Engineering, REST APIs

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Java, Python, JavaScript, SQL
- **Systems & Concurrency:** Multithreading, Thread-Safe Queues, Backpressure & Load Shedding, Worker Pools, Row-Level & Advisory Locking, Idempotency, Caching, Memory-Bounded Processing, Profiling, Graceful Degradation
- **Backend & APIs:** Node.js, Express.js, FastAPI, Flask, REST APIs, Microservices, Async Processing, Server-Sent Events, JWT, OAuth, RBAC, Rate Limiting
- **Databases & Storage:** PostgreSQL, MongoDB, MySQL, SQLite, Redis, S3 | **ORM/ODM:** Mongoose, SQLAlchemy
- **DevOps & Tools:** Docker, Docker Compose, Git, GitHub Actions, CI/CD, Linux, Jira, Postman, pytest, mypy, ruff, $\LaTeX$
- **AI & Data:** LLM Integration, Agentic AI, ReAct Agents, RAG, Model Context Protocol, Prompt Engineering, PyTorch, NumPy, Pandas, Scikit-learn, OpenCV
- **Practices:** System Design, API Design, Code Reviews, TDD, Agile Delivery

KEY SUBJECTS COMPLETED

- Operating Systems, Computer Networks, Computer Architecture, DBMS, Data Structures & Algorithms, Software Engineering, Artificial Intelligence, Machine Learning, Satellite Image Processing, GIS, Remote Sensing

SCHOLASTIC ACHIEVEMENTS

- Secured **GATE (CS)** with **98.94 Percentile** (Mar'25)
- **Finalist**, HackSVIT Hackathon (May'22)
- **Gold Medalist** across **150+** affiliated institutes, Diploma in Information Technology, **Gujarat Technological University (GTU)** (Jan'22)
- **Winner**, Hacktober Hacks Hackathon (Oct'21)
- **Winner**, Hackout'21 (Pre-made Projects) (Sep'21)
- Awarded **Best Student Award**, Diploma in Information Technology, LJ Polytechnic (Nov'19)
- Recipient of **Rajya Puraskar**, the highest state-level honour by **The Bharat Scouts & Guides**, awarded by the Governor of Gujarat (Feb'17)

POSITIONS OF RESPONSIBILITY

- **Teaching Assistant**, IIT Bombay, teaching **30+ postgraduate students**. (Jul'25-Present)
- **Interview Coordinator**, IITB Placement Office: **2,100+ students**, **300+ member team**, **10+ companies**. (Oct'25-Dec'25)
- **Lead Organizer**, HackVGEC 2023, a **state-level hackathon** (Feb'23-Apr'23)
- **Lead Organizer**, TechXIT 2022, an **institute-level technical festival** (Sep'22-Oct'22)